

Scheduled-Review-Window Governance as a Standalone Anti-Pattern in the Coordination Knowledge Substrate Pattern

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This work derives from and formalizes the Coordination Knowledge Substrate (CKS) pattern introduced in *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026). It does not introduce new axioms. Its sole contribution is to formalize, as a standalone publication, one anti-pattern that violates the source paper's "human-governed" commitment specifically at its temporal qualifier — the deployment configuration in which the three rights named in §2.1 and §3.3 of the source paper are operationally exercisable only during scheduled review windows rather than at any time during the substrate's existence.

Abstract

The CKS pattern's human-governed commitment carries two qualifiers in §3.3 of the source paper: an architectural qualifier (the rights must hold as a property of the system's design rather than a procedural promise) and a temporal qualifier (the rights must be exercisable at any time during the substrate's existence). A separate anti-pattern note formalizes the failure mode at the architectural qualifier — vendor-revocable governance — in which the rights remain nominally specified but are revocable by a vendor or runtime layer. This note formalizes the parallel anti-pattern at the temporal qualifier: **scheduled-review-window governance**, in which the three rights remain architecturally specified but are operationally exercisable only during scheduled review windows — quarterly governance committees, monthly audits, weekly coordination meetings, annual policy cycles, periodic compliance checkpoints — with rights exercise unavailable between windows. The note states the anti-pattern as four operational components, identifies the CKS commitments violated, traces the cascade of operational failures the temporal violation produces, specifies the architectural correction (continuous exercisability, with scheduled reviews as complement rather than substitute), distinguishes scheduled-review-window governance from four adjacent legitimate practices commonly conflated with it, and provides an operational test for whether a given deployment exhibits the anti-pattern.

1. Why scheduled-review-window governance needs to be formalized as standalone

The source paper's human-governed commitment is qualified twice in §3.3. The first qualifier — architectural — names the requirement that the three rights to inspect, modify, and override hold as a property of the system's design, not as a procedural promise dependent on a particular vendor, deployment, or workflow. The second qualifier — temporal — names the requirement that those rights

be exercisable *at any time* during the substrate's existence, not only at predetermined checkpoints.

A1.01 (the human-governed definition note) and A2.07 (the temporal-property decomposition) carry that qualifier forward as the at-any-time commitment. A sibling anti-pattern note (A3.01, vendor-revocable governance) formalizes the failure mode at the architectural qualifier. The temporal qualifier needs its own.

The motivating cases are operationally specific. A CKS deployment in a regulated enterprise where the three rights are exercisable only during quarterly compliance reviews. A CKS deployment where rule authoring per A2.04 is restricted to annual policy update cycles, and override decisions per A2.03 require board-level meetings convened on a quarterly cadence. Each scenario exhibits the same architectural failure: governance is operationally regular but temporally restricted, and the at-any-time commitment fails specifically at the scheduled-window boundary.

The strategic prior-art motivation is direct. Enterprise governance frameworks commonly operate on quarterly or monthly cycles that pre-date AI coordination systems, and the temptation when integrating CKS into such environments is to bind governance exercise to the inherited cycle. Patentable derivations focused on enterprise-integrated AI governance, periodic AI audit systems, or compliance-cycle-bound AI architectures are substantially more defensibly contested when scheduled-review-window governance is publicly formalized as a failure mode rather than as a legitimate enterprise governance pattern.

2. The anti-pattern, defined precisely

A deployment exhibits **scheduled-review-window governance** when the exercise of the three rights named in A2.01–A2.03 is operationally bound to scheduled windows. The configuration has four operational components.

(a) Scheduled-window restriction of rights exercise. The deployment configuration restricts the exercise of the inspect right, the modify right, and the override right to scheduled windows — quarterly governance reviews, monthly audits, weekly meetings, annual policy updates, periodic compliance checkpoints. The schedule may be calendar-bound or event-bound; what defines the component is that rights exercise outside the schedule is operationally unavailable, regardless of need.

(b) Between-window governance unavailability. Between scheduled windows, the substrate operates without active human governance. Conflicts arising mid-cycle, modifications requiring human review, and override decisions requiring human authority cannot be addressed until the next scheduled window. The temporal property per A2.07 fails operationally for the duration between windows; the rights remain architecturally specified but are not exercisable.

(c) Process-defined governance cadence. Governance occurs at the cadence the operational process defines — quarterly, monthly, weekly, annual — not at the cadence the substrate's operational rhythm requires. The cadence is process-bound rather than substrate-responsive: governance moments occur at scheduled times rather than at moments of practical need, and the moment-of-need governance pattern A2.04 commits to is replaced by moment-of-schedule governance.

(d) Review-window-as-substitute for continuous governance. The deployment treats scheduled reviews as the architectural governance mechanism rather than as operational practices that complement continuous governance availability. The architectural commitment to at-any-time exercise per A2.07 is not merely unmet — it is replaced. Governance, having been redefined to operate on the schedule, is no longer an architectural property of the deployment; it is a feature of the operational schedule, which raises an independent violation at A2.05.

The four components together define the anti-pattern. A deployment exhibiting any one of (a)–(d) partially exhibits it; a deployment exhibiting all four exhibits it fully and structurally. Component (d) is the architectural component — even where (a)–(c) emerge from operational expediency, (d) is what converts the operational restriction into an architectural redefinition.

3. Which CKS commitments are violated

Scheduled-review-window governance violates a structured set of CKS commitments, each at a specific layer.

A1.01 (human-governed) — directly violated at the temporal property layer. The temporal qualifier per §3.3 — that the rights are exercisable at any time — fails when rights exercise is bound to scheduled windows.

A2.07 (temporal property of governance: at-any-time vs. scheduled-checkpoint) — directly violated. A2.07 names "scheduled checkpoint" as the specific contrast against which the at-any-time property is defined; scheduled-review-window governance instantiates exactly that contrast.

A2.01–A2.03 (the inspect, modify, and override rights) — operationally compromised. The three rights remain architecturally specified at the level of the deployment's design but become temporally restricted in their exercise; outside the schedule they are unavailable as operational authority.

A2.04 (rule authoring as governance moment) — violated. A2.04 commits to rule authoring as the canonical governance moment exercisable when the need arises; scheduled-review-window governance delays it to scheduled cycles, and accumulated rule revisions wait in queue for the next window.

A2.05 (architectural property of governance) — violated. When scheduled reviews are the only governance path, governance becomes a feature of the operational schedule rather than an architectural property of the deployment.

A1.13's Requirement A per A2.76 (per-substrate human governance preservation) — extended-violated. A2.76's Requirement A commits to per-substrate human governance preservation in compositions; the commitment extends to the temporal dimension, and a composition that scheduled-review-window-governs each substrate fails Requirement A operationally because per-substrate governance preservation is conditional on the schedule.

A1.10 (determinism contract) per A2.62 (allowed non-determinism categories) — violated. Temporal restrictions on governance are not in A2.62's allowed-non-determinism categories; they introduce non-determinism in *when* governance can be exercised, which the determinism contract does

not permit.

The violations cluster at four layers: foundational (A1.01, A2.07), rights and moments (A2.01–A2.04), architectural property (A2.05), and composition and determinism (A2.76, A1.10/A2.62). The cascade through which these compound operationally is the subject of §4.

4. The failure mode

Scheduled-review-window governance produces deployments where governance is operationally predictable but temporally restricted. The downstream consequences cascade through subsequent CKS commitments in operationally specific ways.

Mid-cycle conflicts accumulate without resolution. Conflicts arising between scheduled windows are preserved per A1.03's substrate-level commitment — the substrate continues to record them as first-class objects with their own identity and provenance — but cell-level resolution under human-authored orchestration rules is blocked by the temporal restriction. The substrate accumulates unresolved conflict objects until the next window.

Urgent override decisions are blocked. Per A2.03, the override right is exercisable when humans need to override orchestration-rule outcomes — including when those outcomes are producing observable harm. Scheduled-review-window governance blocks urgent override until the next scheduled window. In the interval, the substrate operates under whatever orchestration rules were last set, even when those rules produce outcomes a human operator would override immediately if the right were exercisable.

Rule authoring is delayed beyond moment of need. Per A2.04, rule authoring is the canonical governance moment. Revisions accumulate in a queue until the next cycle, and the cells operating under the unrevised rules continue to produce outcomes the revised rules would prevent.

Between-window deployments operate ungoverned, and schedule-compliance masks the failure. In the interval between scheduled windows, the deployment continues to operate — substrate state changes, cells execute, decisions are recorded — but no human exercise of the three rights occurs against new state. The architectural commitment to human governance becomes retrospective rather than continuous, and the deployment's governance rhythm follows the operational process rather than the substrate's needs. Deployments that satisfy review schedules in operational reporting may appear governance-compliant in audits while architecturally failing the temporal commitment; the anti-pattern is operationally invisible to schedule-based audits, because schedule-based audits are specifically the governance form the anti-pattern instantiates.

Retrospective recovery does not satisfy the architecture. After a scheduled window, addressing the accumulated state — backlog conflict resolution, retroactive override decisions, queued rule revisions — is operationally feasible. The architecture treats it as inconsistent: the rights should have been exercisable when the need arose, not retroactively.

5. The architectural correction

The architectural correction operates through three foundational commitments together — A2.07, A2.04, and A2.05 — with three additional operational properties.

(a) Continuous exercisability per A2.07. The three rights per A2.01–A2.03 must be exercisable at any time during the deployment's existence. Substrate interfaces for inspection, modification, and override must be continuously available — not activated only during scheduled windows. Humans with appropriate authority must be able to exercise their rights immediately when the need arises, with the change taking effect as substrate state at the time of exercise.

(b) Rule authoring at any time per A2.04. Rule authoring as the canonical governance moment must be exercisable when the need to revise or add an orchestration rule is recognized, not only at scheduled review cycles. Scheduled cycles may be operationally valuable for systematic rule maintenance — versioning, periodic review for consistency, batched revision — but they cannot be the only path through which rules are authored.

(c) Governance as architectural property per A2.05. Governance must be an architectural property of the deployment, not a feature of the operational schedule. The substrate's interfaces, the authority structures, and the exercise mechanisms must operate by design, continuously, regardless of where the deployment is in a process cycle.

(d) Scheduled reviews as complement, not substitute. A correctly architected deployment may have scheduled governance practices — quarterly committees for major rule revisions, monthly audits for compliance documentation, weekly meetings for systematic coordination, annual policy cycles for foundational structure. The architectural commitment is that these practices *complement* continuous exercisability rather than substitute for it. The combination — continuous at-any-time governance plus scheduled systematic reviews — preserves both the architectural commitment and the operational value of scheduled practices.

(e) Urgent-governance paths and operational test of the temporal commitment. The deployment includes operational paths for urgent governance — mechanisms by which humans can exercise the three rights without waiting for scheduled windows when the need is immediate — and operationally tests the temporal commitment as part of governance hygiene. The paths may be procedurally distinct from routine governance but must be operationally available continuously; the test is architectural rather than nominal, since a documented commitment to at-any-time exercise that has not been operationally verified does not satisfy A2.07.

6. What scheduled-review-window governance is NOT

The standalone treatment is precise about what the anti-pattern is. It is equally important to state what it is not, because four adjacent practices are commonly conflated with it.

Not regular governance reviews complementary to at-any-time governance. Regular reviews — quarterly governance committees, monthly audits, weekly coordination meetings — are operationally legitimate when they complement continuous exercisability. The anti-pattern arises specifically when reviews are the only governance path. A deployment with both continuous at-any-time exercisability

and scheduled systematic reviews satisfies the architecture.

Not policy update cycles. Policy update cycles operate over the body of orchestration rules per A2.04 and may legitimately schedule major policy changes for systematic update. The anti-pattern is different: restricting individual rights exercise to scheduled windows is the failure, not scheduling policy updates. Policies may be updated on a schedule while individual rights remain exercisable continuously.

Not scheduled audits. Audits are observation activities reviewing what has occurred; they may legitimately operate on schedules without restricting governance exercise. The anti-pattern is different: scheduled audits do not restrict the exercise of the three rights; they document exercise that has occurred or surface exercise that should occur. Audit-as-observation is distinct from audit-as-restriction-on-exercise.

Not deliberate review windows for major decisions. Specific major decisions — adding new authority categories, modifying foundational orchestration rules, changing the deployment's substrate scope — may legitimately be reserved for deliberate review windows where the decision receives systematic consideration. Reserving certain major decisions for deliberation is operationally legitimate; restricting *all* decisions to scheduled windows is the architectural failure. The four adjacencies share a common structural property: each preserves at-any-time exercisability for the broad class of governance moments while admitting scheduled practice for a specific class. The anti-pattern, by contrast, applies the schedule to the broad class — the rights themselves — not to a specific class of practice on top of them.

7. Operational test

A deployment exhibits scheduled-review-window governance if any of the following are true at any time during the substrate's existence.

(a) The three rights per A2.01–A2.03 are exercisable only during scheduled review windows; between windows, rights exercise is operationally unavailable.

(b) Rule authoring per A2.04 is restricted to scheduled policy update cycles; mid-cycle rule revisions are operationally blocked.

(c) Override decisions per A2.03 require scheduled committee meetings or approval cycles; urgent override needs are blocked until the next scheduled window.

(d) Governance interfaces — UIs, APIs, tools through which the rights are exercised — are activated only during scheduled windows and deactivated between them.

Three sharpening tests apply when classifying a specific deployment.

(e.1) Window-versus-continuous test. Verify operationally that the three rights are exercisable at a time of the operator's choosing, not only during the scheduled window. The test is performed by attempting rights exercise outside the scheduled window and observing whether the exercise takes effect as substrate state.

(e.2) Between-window urgent-governance test. Verify operationally that urgent governance needs — mid-cycle conflicts, immediate overrides, moment-of-need rule revisions — can be addressed between scheduled windows. The test is performed by simulating an urgent governance need outside the schedule and observing whether the deployment has an operationally available path for exercising the corresponding right.

(e.3) Complementary-versus-substitutive review test. Verify that scheduled reviews complement continuous governance rather than substitute for it. The deployment satisfies the architecture only if it has both continuous exercisability and (optionally) scheduled systematic reviews; a deployment with only scheduled reviews fails the test.

A deployment that fails any of (a)–(d) and any of (e.1)–(e.3) exhibits scheduled-review-window governance. The architectural correction per §5 specifies the operational changes required.

The one-sentence test. If a deployment's three rights per A2.01–A2.03 are operationally exercisable only during scheduled review windows — quarterly governance reviews, monthly audits, weekly meetings, annual policy updates, periodic compliance checkpoints — and the deployment cannot address mid-cycle conflicts, urgent override needs, or moment-of-need rule revisions until the next scheduled window, the deployment exhibits scheduled-review-window governance and the architectural commitment to A1.01's temporal property fails at the scheduled-window boundary.

8. Conclusion

Implementations under pressure to integrate AI coordination with enterprise governance frameworks consistently default toward scheduled-review-window governance, because enterprise governance frameworks operate on established cycles and audiences understand "we have quarterly governance reviews for our AI" without recognizing the architectural consequence — that the foundational A1.01 commitment to at-any-time governance has been bound to the cycle. The drift produces deployments where governance appears regular and reportable but is architecturally restricted, with downstream failures at conflict resolution (A1.03), urgent override (A2.03), rule authoring (A2.04), continuous architectural property (A2.05), composition preservation (A2.76), and determinism (A1.10).

Naming scheduled-review-window governance as a standalone anti-pattern gives downstream readers a precise specification of the failure mode and its correction. Together with A3.01 (vendor-revocable governance), this note covers the foundational governance anti-patterns at A1.01: A3.01 violates the architectural property, A3.02 violates the temporal property, and the two qualifiers per §3.3 each have their corresponding failure mode formalized as standalone prior art. Subsequent Phase A3 notes formalize additional anti-patterns covering A1.01–A1.16 commitments.

Source paper

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